

Real MSPT gains (ship profile)

Prove YaP-Folia **ship knobs** beat stock Folia and rank ahead of Canvas — knobs **disclosed** in result JSON (`knob_*`).

Truth table (2026-09-04)

Scenario	Stamp	Outcome
fullcite ship (peak)	20260904TshipFc2	YaP −12.40% vs stock — citeable ; knobs proven
fullcite re-verify	20260904T040935Z	YaP −5.53% vs stock — citeable ship gate PASS (knobs disclosed)
fullcite 3-way	20260904TshipFc	YaP #1 vs Folia & Canvas (Canvas ~35 mspt); pairwise −4.8% that run (noise)
heavypop ship	20260904Tship0n	YaP −7.55% vs stock; #1 vs Canvas (−3.85%)
heavypop baseline	20260904TshipBase	YaP −7.19% (async+hopper only; smart knobs off)

Ship cite profile

Knob	Value
async-chunk-save	true
hopper-tick-budget	64
entity-tick-budget	400 (MSPT-gated @ 12)
microtick-budget-ms	8
subregion-partition	true

Never defer players / TNT / vehicles / items / bosses. `fuse_ticking_ok` required. Gate: `YAP_MSPT_REQUIRE_SHIP_KNOBS=1` .

Why micro / subregion / parallel

1. **Smart entity budget + microtick** — soft-cap Mob AI when a region is hot (≥ 12 ms MSPT)
2. **Subregion partition** — parallel Folia shards when hot + geometry allows

Product features, disclosed on every cite. Baseline A/B (YAP_BENCH_CITE_BASELINE=1) shows async+hopper alone (~-7% heavypop); ship profile adds headroom under hot fullcite.

Stability

```
./scripts/yapctl soak-compatible # PASS 20260904T033554Z (ship knobs ON)
./scripts/yapctl soak-perf 30 # PASS 20260904T033626Z
./scripts/yapctl cite-fullcite # PASS 20260904T040935Z (-5.53%; peak cite -12.40% at shipFc2)
./scripts/yapctl soak-long 12 # background uptime; not a cite gate
```

Anti-gaming

- Disclose knobs; same load proofs; no keepalive-only population cites
- Do not claim “vanilla Folia settings” when ship smart knobs are on